Link Path Network Optimization Report

Model Type: **link_path**

Metric	Value
Status	Optimal
Objective Value	$830.00
Total Revenue	$830.00
Total Expansion Cost	$0.00
Total Installation Cost	$0.00
Total Cost	$0.00
Net Profit	$830.00
Accepted Demands	6 / 6
Rejected Demands	0 / 6
Acceptance Rate	100.0%
Nodes	8
Edges	12
Total Initial Capacity	280.00
Total Expansion	0.00
Capacity Increase (%)	0.0%

Insights & Highlights

Key Insight	Detail
Most Congested Edge	S2 -> R3 (100.0%)
Number of Highly Utilized Edges (≥90%)	4
Most Capacity Expansion	None
Total Lost Revenue	$0.00
Number of Expanded Edges	0

WARNING: Congestion risk in the following edges (utilization ≥90%):
- Edge S2 -> R3: 100.0% used
- Edge R1 -> R3: 100.0% used
- Edge R3 -> D1: 90.0% used
- Edge R3 -> D2: 100.0% used

Accepted Demands & Flow Paths

Demand 0: S1 -> D1

Bandwidth: 10.0 units **Revenue:** $150.0
Revenue/BW: $15.00/unit

Path	Fraction	Bandwidth
S1 -> R1 -> R3 -> D1	20.00%	2.00
S1 -> R2 -> R3 -> D1	80.00%	8.00

Demand 1: S1 -> D2

Bandwidth: 8.0 units **Revenue:** $120.0
Revenue/BW: $15.00/unit

Path	Fraction	Bandwidth
S1 -> R1 -> R3 -> D2	100.00%	8.00

Demand 2: S2 -> D1

Bandwidth: 12.0 units **Revenue:** $180.0
Revenue/BW: $15.00/unit

Path	Fraction	Bandwidth
S2 -> R3 -> D1	100.00%	12.00

Demand 3: S2 -> D2

Bandwidth: 15.0 units **Revenue:** $200.0
Revenue/BW: $13.33/unit

Path	Fraction	Bandwidth
S2 -> R3 -> D2	6.67%	1.00
S2 -> R2 -> R3 -> D2	60.00%	9.00
S2 -> R2 -> R4 -> D2	33.33%	5.00

Demand 4: S1 -> D1

Bandwidth: 5.0 units **Revenue:** $80.0
Revenue/BW: $16.00/unit

Path	Fraction	Bandwidth
S1 -> R1 -> R3 -> D1	100.00%	5.00

Demand 5: S2 -> D2

Bandwidth: 7.0 units **Revenue:** $100.0
Revenue/BW: $14.29/unit

Path	Fraction	Bandwidth
S2 -> R3 -> D2	100.00%	7.00

Rejected Demands

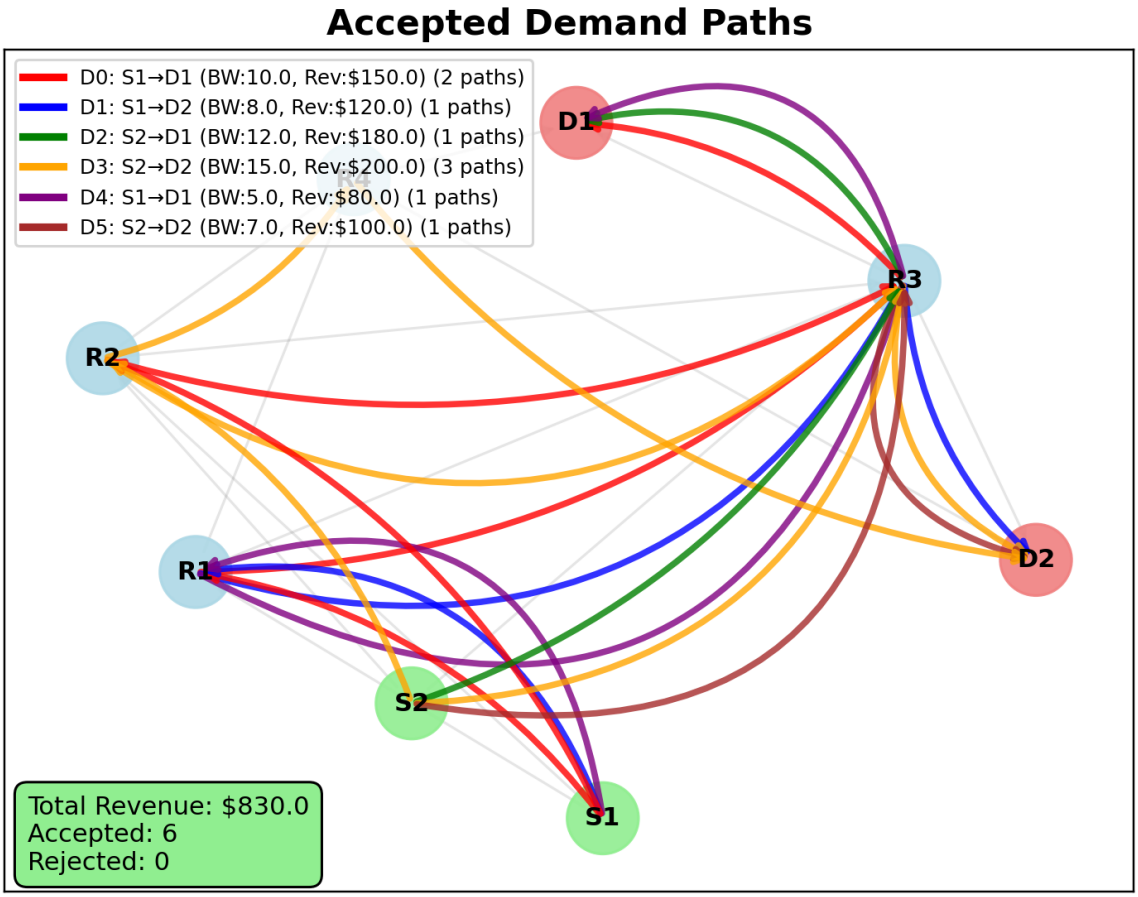
No demands were rejected.

Edge Utilizations (Top 12 Edges)

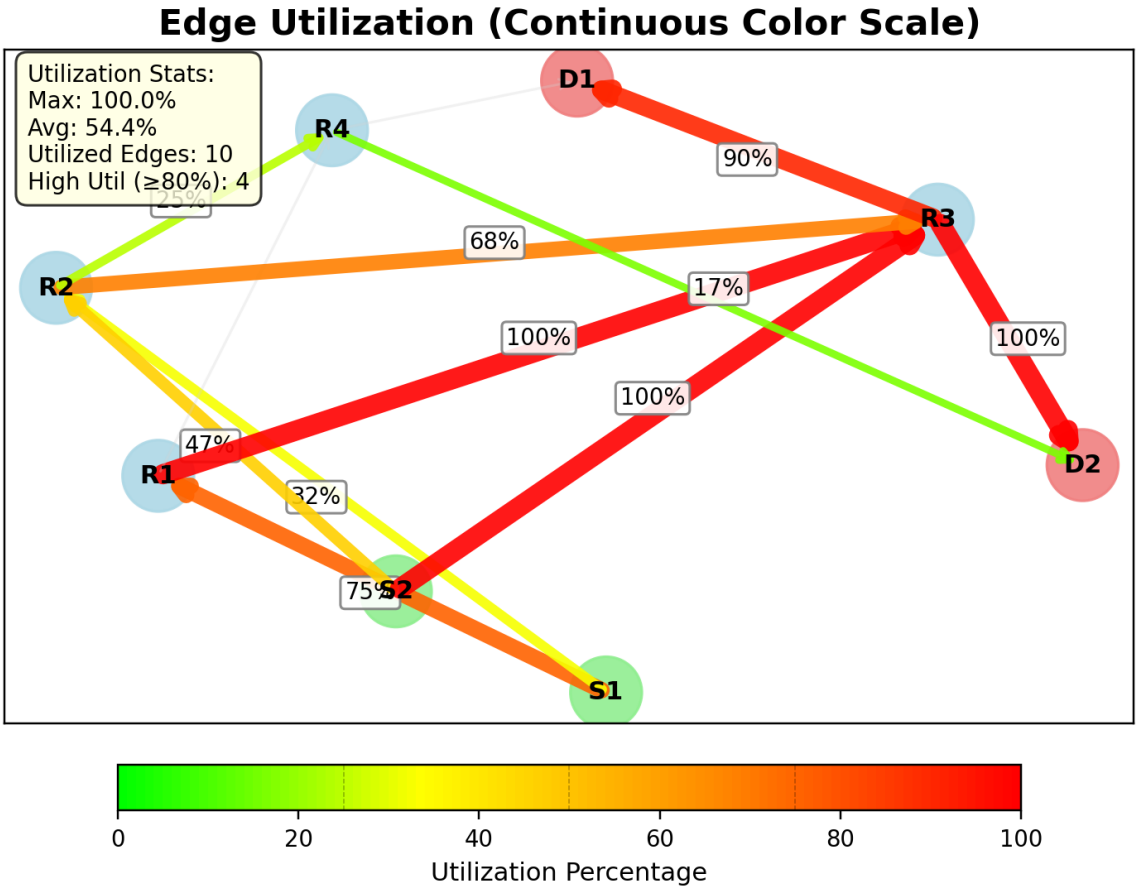
Edge	Flow	Total Capacity	Utilization (%)
S2 -> R3	20.00	20.00	100.0 %
R3 -> D2	25.00	25.00	100.0 %
R1 -> R3	15.00	15.00	100.0 %
R3 -> D1	27.00	30.00	90.0 %
S1 -> R1	15.00	20.00	75.0 %
R2 -> R3	17.00	25.00	68.0 %
S2 -> R2	14.00	30.00	46.7 %
S1 -> R2	8.00	25.00	32.0 %
R2 -> R4	5.00	20.00	25.0 %
R4 -> D2	5.00	30.00	16.7 %
R1 -> R4	0.00	20.00	0.0 %
R4 -> D1	0.00	20.00	0.0 %

Network Analysis Visualizations

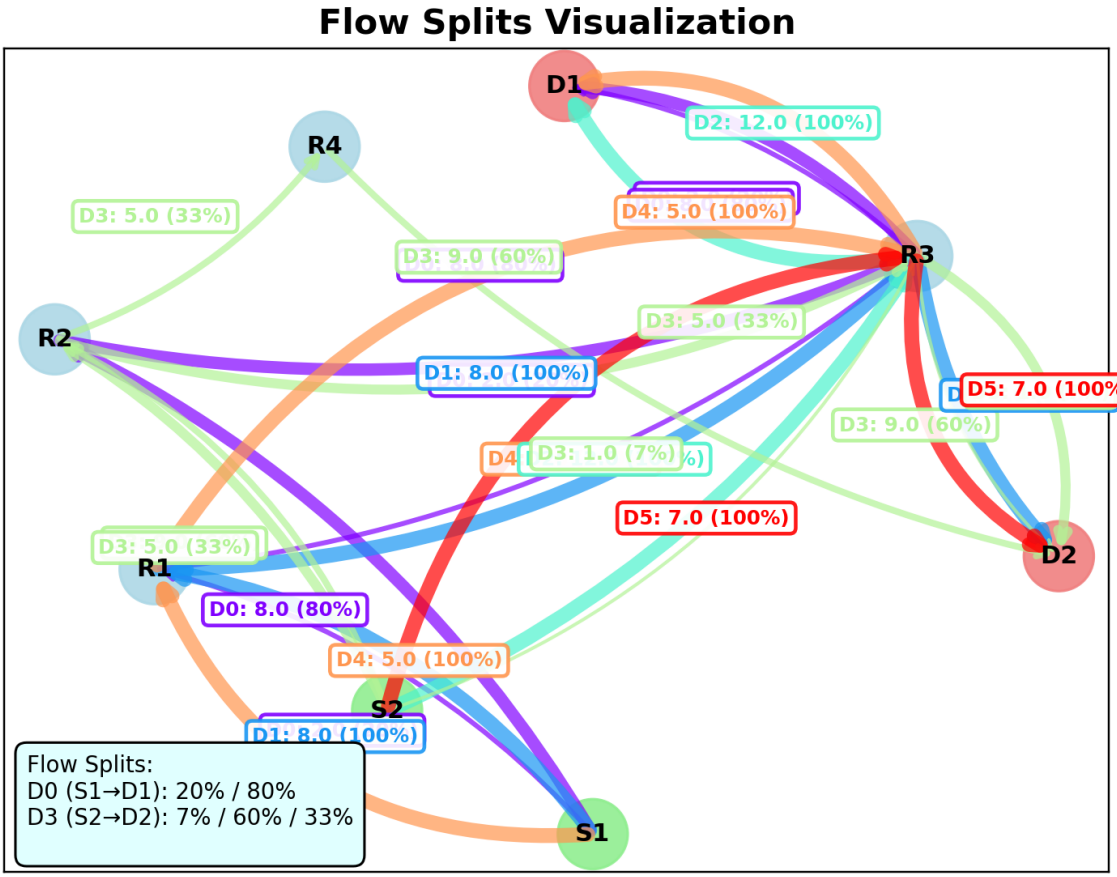
Accepted Demand Paths



Edge Utilization



Flow Splits


Flow Splits Visualization
D1
D2: 12.0 (100%)
R4
D3: 5.0 (33%)
D4: 5.0 (100%)
D3: 9.0 (60%)
R3
R2
D3: 5.0 (33%)
D1: 8.0 (100%)
D5: 7.0 (100%)
D3: 9.0 (60%)
D3: 1.0 (7%)
D5: 7.0 (100%)
D3: 5.0 (33%)
R1
D2
D0: 8.0 (80%)
D4: 5.0 (100%)
S2
D1: 8.0 (100%)
Flow Splits:
D0 (S1→D1): 20% / 80%
D3 (S2→D2): 7% / 60% / 33%
S1